Computer Networks Spring 2026

Assignment#1 (6A & 6B) Solution

Part – 1

Review Questions:

12. A circuit-switched network can guarantee a certain amount of end-to-end bandwidth for the duration of a call. Most packet-switched networks today (including the Internet) cannot make any end-to-end guarantees for bandwidth. FDM requires sophisticated analog hardware to shift signal into appropriate frequency bands.

R19.

(1) Throughput = min (R1, R2, R3) = min(500, 2000, 1000) kbps = 500 kbps.

(2) Time = bits/throughput = (8 * 4,000,000) bits/500,000 bps = 64 seconds

(3) Now, R2 = 100 kbps

Throughput = min(R1, R2, R3) = min(500, 100, 1000) kbps = 100 kbps

Time = bits/throughput = (8 * 4,000,000) bits/100,000 bps = 320 seconds

Problems:

Problem 3

a) A circuit-switched network would be well suited to the application, because the application involves long sessions with predictable smooth bandwidth requirements. Since the transmission rate is known and not bursty, bandwidth can be reserved for each application session without significant waste. In addition, the overhead costs of setting up and tearing down connections are amortized over the lengthy duration of a typical application session.

b) In the worst case, all the applications simultaneously transmit over one or more network links. However, since each link has sufficient bandwidth to handle the sum of all of the applications' data rates, no congestion (very little queuing) will occur. Given such generous link capacities, the network does not need congestion control mechanisms.

Problem 6

a) $d_{prop} = m / s$ seconds.

b) $d_{trans} = L / R$ seconds.

c) $d_{end-to-end} = (m / s + L / R)$ seconds.

d) The bit is just leaving Host A.

e) The first bit is in the link and has not reached Host B.

f) The first bit has reached Host B.

g) We want $\boldsymbol{m = \frac{L}{R}s = \frac{1500\times8}{10\times10^6}(2.5\times10^8) = 3\times10^5 = 300}$ km.

Problem 10

The first end system requires L/R_1 to transmit the packet onto the first link; the packet propagates over the first link in d_1/s_1; the packet switch adds a processing delay of d_{proc}; after receiving the entire packet, the packet switch connecting the first and the second link requires L/R_2 to transmit the packet onto the second link; the packet propagates over the second link in d_2/s_2. Similarly, we can find the delay caused by the second switch and the third link: L/R_3, d_{proc}, and d_3/s_3.
Adding these five delays gives

$$d_{end\text{-}end} = L/R_1 + L/R_2 + L/R_3 + d_1/s_1 + d_2/s_2 + d_3/s_3 + d_{proc} + d_{proc}$$

To answer the second question, we simply plug the values into the equation to get 4.8 + 4.8 + 4.8 + 20+16 + 4 + 3 + 3 = 60.4 msec.

Part – 2

Question: 01

Calculation of Constant Delays

- **Processing Delay:** 2 ms (given in original problem).
- **Transmission Delay**
  - At 100% rate: 5000 bits/ (10 x 10^6 bps) = 0.5 ms
  - At 75% rate (for Part B): 5000 bits/ (7.5 x 10^6 bps) = 0.67 ms (rounded for calculation).
- **Propagation Delay:** 400 x10^3 m/ (3 x 10^8 m/s = 1.33 ms (rounded for calculation).

(1a): Packets 1 to 5 (Original Rate)

At T0=0, R1 begins processing. A packet **leaves** a node after d_proc + d_trans and **arrives** at the next node after adding d_prop.

- **T1 (Leaves R1):** d_proc + d_trans = 2 + 0.5 = 2.5 ms.
- **T2 (Arrives at R2):** T1 + d_prop = T1 + 1.33 ms.
- **T3 (Leaves R2):** T2 + d_proc + d_trans = T2 + 2.5 ms.
- **T4 (Arrives at R3):** T3 + d_prop = T3 + 1.33 ms.

Packet#	T1	T2	T3	T4
1	2.5	3.83	6.33	7.66
2	5.0	6.33	8.83	10.16
3	7.5	8.83	11.33	12.66
4	10.0	11.33	13.83	15.16
5	12.5	13.83	16.33	17.66

(1b): Packets 6 to 10 (75% Transmission Rate)

The transmission rate drops to 7.5 Mbps, so d_trans becomes 0.67 ms. The processing delay remains 2 ms.

- **T1 (Leaves R1):** Packet 6 leaves d_proc + d_trans_new after Packet 5.
- Calculations follow the same logic as above using the new d_trans = 0.67 ms.

Packet	T1	T2	T3	T4
6	15.17	16.5	19.17	20.5
7	17.84	19.17	21.84	23.17
8	20.51	21.84	24.51	25.84
9	23.18	24.51	27.18	28.51
10	25.85	27.18	29.85	31.18

(1c): Delay Comparison

To find the total delay for a single packet (from T_0 at R1 to arrival at R3):

1. **Original Case (100%):** T4 of Packet 1 = 7.66 ms.
2. **New Case (75%):** Using the logic for a single packet starting at T_0:
   - T1 = 2 + 0.67 = 2.67 ms
   - T2 = 2.67 + 1.33 = 4.00 ms
   - T3 = 4.00 + 2 + 0.67 = 6.67 ms
   - T4 = 6.67 + 1.33 = 8.00 ms

Percentage Increase:

$$(8.00 - 7.66)/7.66 \times 100 = 4.44\%$$

The total delay for a single packet increases by approximately 4.44% when the transmission rate is reduced to 75%.

Question 2:

Bandwidth is potential capacity; Throughput is actual delivered bits/sec. Scenario: You have a 1 Gbps fiber connection (High Bandwidth), but you are downloading from a server that only has a 1 Mbps upload connection. Your throughput is capped at 1 Mbps.

Question 3:

Case A: N_cs = 2,000,000 / 200,000 = 10 users.

Case B:

1. Binomial setup: $\binom{30}{15}(0.1)^{15}(0.9)^{15}$

2. Because users are idle 90% of the time, 10 users rarely transmit at the exact same moment. Packet switching gambling on this probability allows over-provisioning (admitting 30 users on a link that only supports 10 peak users).

Question 4:

a. **Available Bandwidth:**
   - Backbone: 100 Gbps / 50,000 = 2 Mbps.
   - ISP Core: 10 Gbps / 2,000 = 5 Mbps.

b. **Bottleneck Identification:**
   - Server: 1 Gbps (Plenty)
   - Backbone per flow: 2 Mbps
   - ISP per flow: 5 Mbps
   - Home: 5 Mbps. The lowest per-flow rate is the Backbone (2 Mbps).

c. **Throughput:** Max theoretical throughput = **2 Mbps**.